

Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603 110

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Department of Computer Science and Engineering

Assignment – II
Regulations – R2021

Degree & Branch	B.E. & CSE			Semester	VI
Subject Code & Name	UCS2603 & Theory of Computation				
Academic Year	2024-2025 EVEN	Batch	2022- 2026	Given date: 28.03.2025	
Due date: 30.03.2025	Answer All Questions			Maximum: 40 marks	

(K1: Remembering, K2: Understanding, K3: Applying, K4: Analyzing, K5: Evaluating)

CO1	Construct mathematical proofs related to computation and finite automata for a given language (K3)
CO2	Identify relationship between finite automata and regular expressions (K3)
CO3	Distinguish different types of grammars, analyse relationship between language and context free grammar and construct PDA for CFG (K4)
CO4	Construct Turing machine for a given language (K3)
CO5	Explain the decidability or undecidability of various problems and make use of it (K3)

1	Design a Turing Machine in inventory management for a warehouse system. Suppose a warehouse keeps track of the stock of products in unary notation. When a product is sold, the quantity of the product in the warehouse needs to be updated. The Turing machine can perform subtraction between the current stock (m) and the quantity sold (n), ensuring that the stock cannot become negative. If the quantity sold exceeds the current stock, the Turing machine will replace the result with the Blank (B) symbol, indicating zero stock and preventing negative values. This ensures that the system maintains accurate, non-negative stock levels always (K6, CO4).	1.4.1, 2.1.3, 2.4.1, 13.3.1
2	Design a 3-track Turing Machine to solve the following problem. Given a number “n” in unary (represented by a sequence of n 1's), the task is to determine if “n” is a prime number. A prime number has no divisors other than 1 and itself. If the number is divisible by any other number between 2 and n–1, it is not prime (K6, CO4).	1.3.1, 1.4.1, 10.3.1, 13.3.1
3	Explain the distinction between a language that is Recursive (R), Recursively Enumerable (RE), not Recursively Enumerable (RE) and a language that is undecidable but RE. Provide an example of each, referencing undecidable	1.3.1, 1.4.1, 2.1.3, 2.4.1

	problems in the context of Turing Machines, and discuss their potential applications in real-world computational systems (K5, CO5).	
4	In distributed systems, databases, or file synchronization tools, ensuring that multiple copies of data remain consistent is critical. Assume two versions of a file or database and need to be verified for the possibility of consistent state using an appropriate approach for the given input sequences. List A represents a sequence of operations applied to the first version of the data. List B represents a sequence of operations applied to the second version of the data. List A: 10, 101, 011, 0, 110, List B: 101, 01, 1, 01, 110 Choose the technique to find all possible solution for these lists, it indicates that there's a sequence of operations that can bring both versions of the data to the same state. (K5, CO5).	1.3.1, 1.4.1, 13.3.1